Introduction & Background

As organizations increasingly rely on data analytics to enhance their engagement strategies for various audiences, social media has become a major catalyst in marketing and audience interactions. With millions of posts seeking attention across Social Media platforms, it has become an art and science to understand how content has gone viral. This project focuses on examining that phenomenon through a data-driven approach.

Leveraging a rich dataset of social media post metrics, we will research the variables behind successful content that include timing, content category, sentiment and emotional tone, that contribute to high engagement. The overarching goal of this research is to leverage insights for content strategists and marketers, in optimizing the effectiveness of their publishing calendar, creating greater narrative engagement, and strategically utilizing emotions for viral consumption.

We will be using Tableau as our primary tool to facilitate visualization of the dataset and storytelling, while merging calculated fields and custom logic to provide meaningful dashboards. To promote clarity and effectiveness, we will additionally include AI-generated recommendations and peer feedback from the Tableau Community to help us revise visual encodings, layout decisions, and employ different color schemes (Open AI, 2025)

Dataset Overview and Preparation

The dataset used for this project is the Social Media Engagement Dataset uploaded by a user named SubashMaster0411 (Kaggle, 2024), which was originally sourced from Kaggle. The dataset contains thousands of records of social media posts, as well as important KPIs (key performance indicators) which often companies do not take into consideration, however they would offer critical insight about audience engagement and content performance on social media or on their website. Thanks to the organization of the dataset, it is suited for both categorical and time-series analysis, which will really promote more in-depth analysis about engagement of social media content.

To support deeper insight and enhance visual storytelling, we applied several key transformations and enhancements to the dataset. We calculated additional fields such as posting hour, time of day, posting weekday, engagement rate, total engagement, country, city, and dominant emotion, which are helpful for understanding engagement and to examine audience behavior over time, geographical location, and emotional tone. Sentiment values have been cleaned and standardized to ensure consistent values across platforms. We also merged visualizations from different Tableau workbooks into one integrated Tableau file for better filtering and storytelling on the dashboard storytelling. These preparation steps allowed us to have a reliable analysis-ready dataset for providing a clear insight into what drives engagement.

Research Questions

For our analysis and dashboard storytelling, we focused on the 8 core research questions.

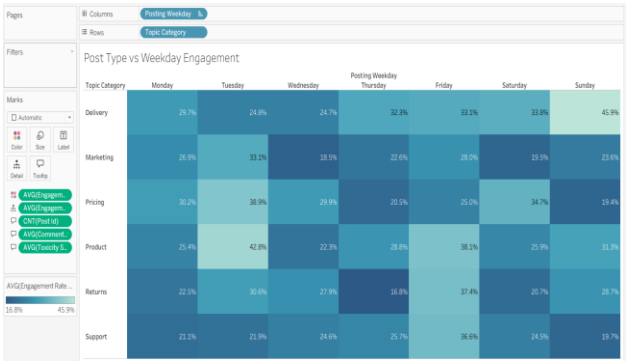
1. How does the average engagement rate vary across content categories and weekdays?
2. How does engagement change over time by sentiment?

3. At what time of day are users most engaged?
4. Do certain emotions lead to higher virality on social media?
5. How does engagement performance differ across campaign phases (Pre-Launch, Launch, Post-Launch)?
6. How do sentiment levels vary across different product brands and social media platforms?
7. Do emotions impact engagement rates across different campaigns?
8. Do engagement rates vary significantly across countries?

Visualization and Insights

Chart Type:	Heatmap
Research Question:	How does average engagement rate vary across content categories and weekdays?
Fields Used:	• **Calculated Field:** ➢ *Engagement Rate (Calculated)* = (Likes Count + Comments Count + Shares Count) / Impressions ➢ *Posting Weekday = DATENAME('weekday', [Timestamp])* • **Placement on Rows and Columns:** ➢ *Topic Category* (Rows), *Posting Weekday* (Columns) • **Marks / Encodings:** ➢ *AVG(Engagement Rate (Calculated) – Color and Label (percentage, 1 decimal place)* ➢ *CNT(Post ID), AVG(Comment Count), AVG(Toxicity Score) - Tooltip*
Enhancements Made:	• Adjusted the solid blue color scale to a diverging palette around the average engagement rate. • The new palette increased the visual ability to pick out engagement outliers and patterns across weekdays and types of posts.
Business Insight:	The heatmap illustrates distinct posting patterns based on their category and weekday. For example, 'Product' posts spike mid-week (Tuesday) and 'Delivery' does best on weekends (particularly Sunday). It is helpful for marketers to have these insights as they can better arrange their publishing, as well as understand that specific days of the week consistently outperform others.

Initial Viz



Final Viz

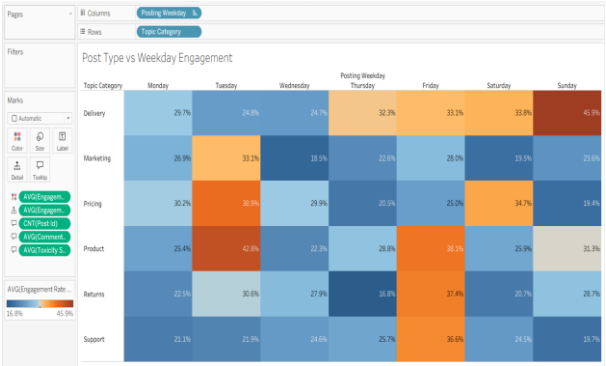
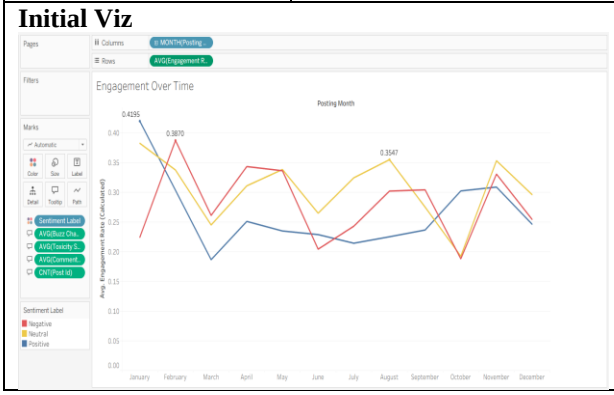


Chart Type:	Time Series Line Chart
Research Question:	How does engagement change over time by sentiment?
Fields Used:	• **Calculated Field:** ➢ *Posting Month = DATETRUNC('month', [Timestamp])* • **Placement on Rows and Columns:** ➢ *Sentiment Label (Rows), Posting Month (Columns)* • **Marks / Encodings:** ➢ *Sentiment Label – Color* ➢ *AVG(Buzz Change Rate), CNT(Post ID), AVG(Comment Count), AVG(Toxicity Score) - Tooltip*
Enhancements Made:	• Added Trend line for each Sentiment Category (Negative, Neutral and Positive) • Adjusted the line thickness and axis scaling for better clarity • The maximum and minimum values were marked.
Business Insight:	The line graph illustrates positive sentiment showed strong engagement. However, when a month featured high negative sentiment, interaction levels also increased sharply. Therefore, content connected to real-world events, or emotionally charged content, may initiate audience engagement spikes, which are only short term.

Initial Viz



Final Viz

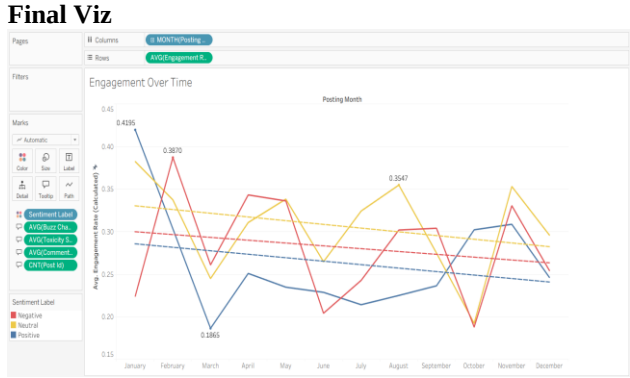
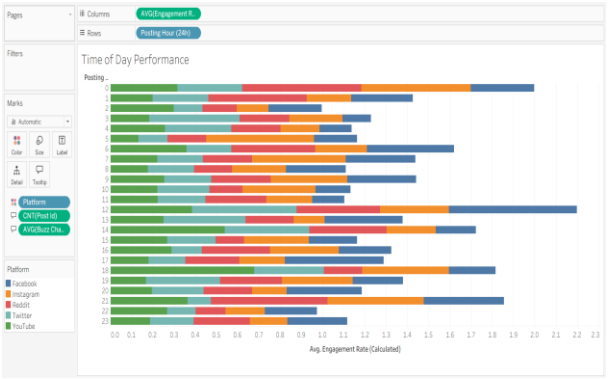


Chart Type:	Horizontal Stacked Bar Chart
Research Question:	At what time of day are users most engaged?
Fields Used:	• **Calculated Field:** ➢ *Engagement Rate (Calculated) = (Likes Count + Comments Count + Shares Count) / Impressions* ➢ *Posting Hour (24h) = DATEPART('hour', [Timestamp])* ***[Used for Enhancement in the Final Viz]*** • *Time of Day = IF DATEPART('hour', [Timestamp]) >= 5 AND DATEPART('hour', [Timestamp]) < 12 THEN "Morning"* ○ *ELSEIF DATEPART('hour', [Timestamp]) >= 12 AND DATEPART('hour', [Timestamp]) < 17 THEN "Afternoon"* ○ *ELSEIF DATEPART('hour', [Timestamp]) >= 17 AND DATEPART('hour', [Timestamp]) < 21 THEN "Evening"* ○ *ELSE "Night"* • **Placement on Rows and Columns:** ➢ *Posting Hour (24h) (Rows), AVG(Engagement Rate (Calculated)) (Columns)* • **Marks / Encodings:** ➢ *Platform – Color* ➢ *AVG(Buzz Change Rate), CNT(Post ID)- Tooltip*

Enhancements Made:	• Replaced the Posting Hour (24h) with Time of Day by grouping hours into morning, afternoon, evening and night for better clarity. • Added Platform as a filter to compare Facebook, Instagram, Twitter and Reddit. • AVG(Engagement Rate (Calculated) was added as a Label (percentage, 1 decimal place) to show the engagement rate percentage on the bars for easy comparison.
Business Insight:	The bar chart illustrates engagement peaks for different platforms. Facebook and Instagram had the highest engagement in late evenings (6–9 PM), while Twitter had overall great engagement throughout the day. This analysis gives us an insight into best practices for posting. Posts that display a high total engagement and a high engagement rate demonstrate the ideal kind of post. It did not just reach people, it also got people to engage with the post. This could eliminate some of the guesswork marketing teams have about what type of post work better on which platform.

Initial Viz



Final Viz

Chart Type:	Box-and-Whisker Plot
Research Question:	How do sentiment levels vary across different product brands and social media platforms?
Fields Used:	**Placement on Rows and Columns:** • Columns: Platform • Rows: Sentiment Score **Marks / Encodings:** • Mark Type: Box-and-Whisker Plot (added via Analytics pane) • Underlying Marks: Circles showing individual sentiment values • Reference Line: Overall mean sentiment score • Color: Neutral palette for clarity • Tooltip: Sentiment Score, Median, Quartiles, Platform
Enhancements Made:	• Converted SUM(Sentiment Score) to raw values to enable boxplot distribution • Reduced dot opacity to prevent overlapping noise • Added an overall mean reference line to support platform comparison • Synced axis range across platforms (−1 to +1) • Cleaned axis labels and formatted tooltips for clarity
Business Insight:	Twitter and Reddit can be seen to have the widest sentiment spread and the most negative outliers, which means that the most polarizing discussions occur there. Facebook and Instagram have more narrow distribution centered towards neutral. Brands should monitor Reddit and Twitter more actively for spikes in sentiment.

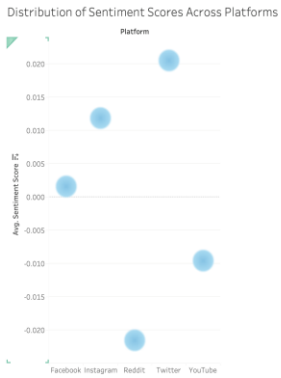
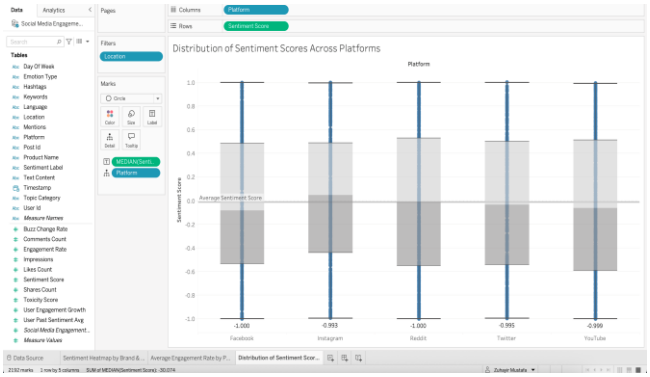
Initial Viz	**Final Viz**
	

Chart Type:	Heatmap
Research Question:	How do sentiment levels vary across different product brands and social media platforms?
Fields Used:	**Placement on Rows and Columns:** • Columns: Platforms, Rows: Brand Name **Marks / Encodings:** • Mark Type: Square • Color: Diverging palette centered around zero to highlight positive vs. negative sentiment • Label: Display average sentiment score on each square • Tooltip: Brand Name, Platform, AVG(Sentiment Score)
Enhancements Made:	• Adjusted color scale to a diverging red–blue palette for clearer sentiment contrast • Added sentiment labels to each cell to improve interpretability • Increased square size and boosted text visibility • Sorted platforms and brand names for cleaner comparison • Added brand filter for easier drill-down
Business Insight:	The chart shows that sentiment is highly platform dependent. Brands like Google and Microsoft are consistently positive compared to Pepsi or Nike, which have negative sentiment or Reddit and Twitter. Campaigns for sensitive brands should prioritize platforms where sentiment has skewed towards a negative bias.

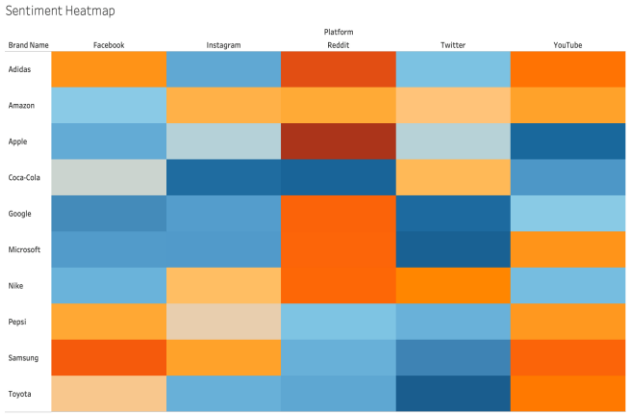
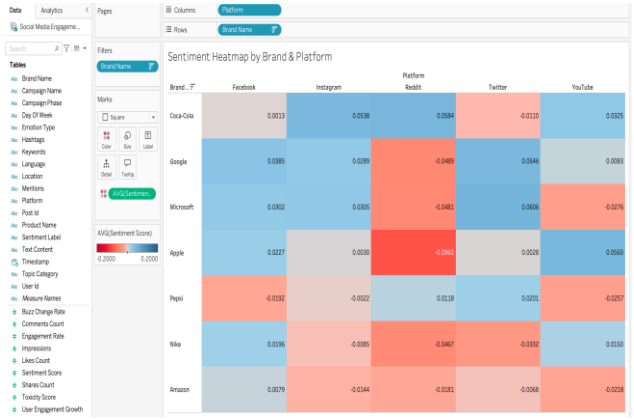
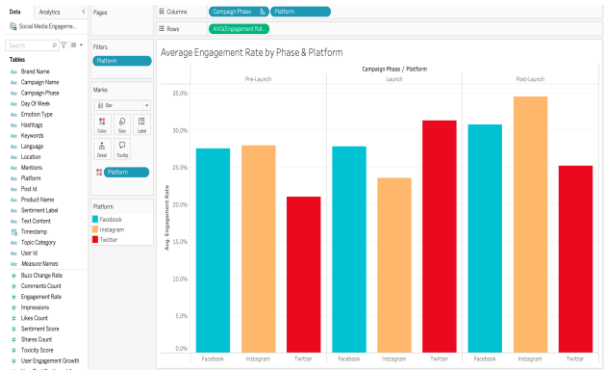
Initial Viz	**Final Viz**
	

Chart Type:	Grouped Bar Chart
Research Question:	How does engagement performance differ across campaign phases (Pre-Launch, Launch, Post-Launch)?
Fields Used:	**Placement on Rows and Columns:** • Columns: Campaign Phase → Platform • Rows: AVG(Engagement Rate) **Marks / Encodings:** • Marks: Bar • Color: Distinct Color per Platform • Tooltip: Platform, Campaign Phase , AVG Engagement Rate • Platform name Filters: Facebook, Instagram, Youtube
Enhancements Made:	• Converted stacked bars to grouped/clustered bars for better platform comparisons • Applied platform-specific color scheme for differentiation • Added gridlines and percentage formatting for readability • Filter added for platform to allow focused analysis • Increased label size and added consistent axis scaling
Business Insight:	Engagement tends to be higher during the Launch and Post Launch stages, which is more evident on Facebook and Instagram. On Twitter the engagement is high during launch but dies down during post-launch. Campaigns should focus on Instagram and Facebook during the final promotional push and put refreshing content on Twitter post launch.

Initial Viz



Final Viz

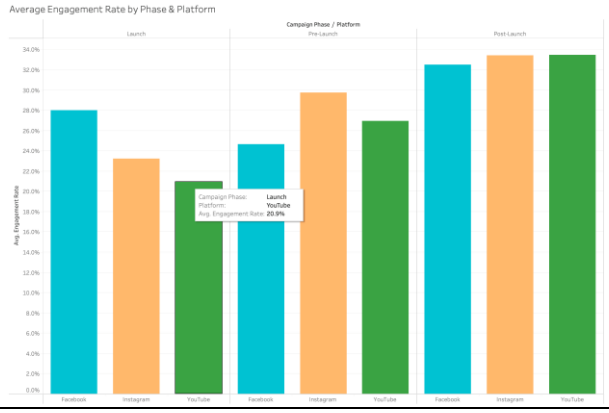


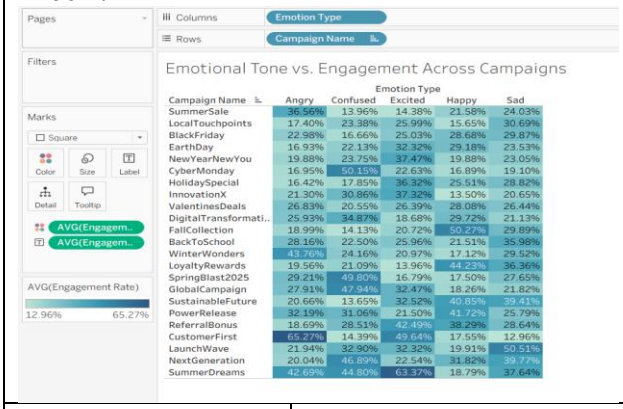
Chart Type:	Heatmap
Research Question:	Do emotions impact engagement rates across different campaigns?

Fields Used:	• **Calculated Field:**
	Engagement Rate = (Likes Count + Comments Count + Shares Count) / Impressions
	• **Rows and Columns:**
	  ○ Columns: Emotion Type
	  ○ Rows: Campaign Name
	
	Marks / Encodings:
	• Color: AVG(Engagement Rate) using a diverging color palette
	• Label: AVG(Engagement Rate) %

Enhancements Made:	• Changed color scheme from blue to diverging (orange/gray) to show sentiment extremes • Added % engagement labels inside cells

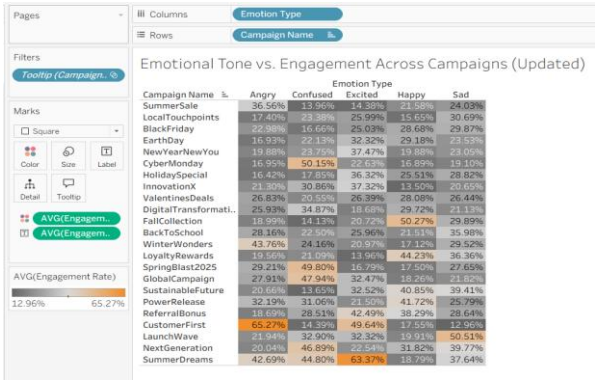
	• Sorted campaigns and emotions for improved readability • Added filters for Campaign Phase or Platform
Business Insight:	The heatmap shows that emotions like "Happy" and "Excited" drive higher engagement across campaigns like BackToSchoolBlitz and NewProductReveal. Meanwhile, emotions like "Sad" or "Angry" consistently underperform. Understanding which emotion drives engagement helps tailor campaign tone accordingly.

Initial Viz



Campaign Name	Angry	Confused	Excited	Happy	Sad
SummerSale	36.56%	13.96%	14.38%	21.58%	24.03%
LocalTouchpoints	17.40%	23.38%	25.99%	15.65%	30.69%
BlackFriday	22.98%	16.66%	25.03%	28.68%	29.87%
EarthDay	16.93%	22.13%	32.32%	29.18%	23.53%
NewYearNewYou	19.88%	23.75%	37.47%	19.88%	23.05%
CyberMonday	16.95%	50.15%	22.63%	16.89%	19.10%
HolidaySpecial	16.42%	17.85%	36.32%	25.51%	28.82%
InnovationX	21.30%	30.86%	37.32%	13.50%	20.65%
ValentinesDeals	26.83%	20.55%	26.39%	28.08%	26.44%
DigitalTransformati..	25.93%	34.87%	18.68%	29.72%	21.13%
FallCollection	18.99%	14.13%	20.72%	50.27%	29.89%
BackToSchool	28.16%	22.50%	25.96%	21.51%	35.98%
WinterWonders	43.76%	24.16%	20.97%	17.12%	29.52%
LoyaltyRewards	19.56%	21.09%	13.96%	44.23%	36.36%
SpringBlast2025	29.21%	49.80%	16.79%	17.50%	27.65%
GlobalCampaign	27.91%	47.94%	32.47%	18.26%	21.82%
SustainableFuture	20.66%	13.65%	32.52%	40.85%	39.41%
PowerRelease	32.19%	31.06%	21.50%	41.72%	25.79%
ReferralBonus	18.69%	28.51%	42.49%	38.29%	28.64%
CustomerFirst	65.27%	14.39%	49.64%	17.55%	12.96%
LaunchWave	21.94%	32.90%	32.32%	19.91%	50.51%
NextGeneration	20.04%	46.89%	22.54%	31.82%	39.77%
SummerDreams	42.69%	44.80%	63.37%	18.79%	37.64%

Final Viz



Campaign Name	Angry	Confused	Excited	Happy	Sad
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LocalTouchpoints	17.40%	23.38%	25.99%	15.65%	30.69%
BlackFriday	22.98%	16.66%	25.03%	28.68%	29.87%
EarthDay	16.93%	22.13%	32.32%	29.18%	23.53%
NewYearNewYou	19.88%	23.75%	37.47%	19.88%	23.05%
CyberMonday	16.95%	50.15%	22.63%	16.89%	19.10%
HolidaySpecial	16.42%	17.85%	36.32%	25.51%	28.82%
InnovationX	21.30%	30.86%	37.32%	13.50%	20.65%
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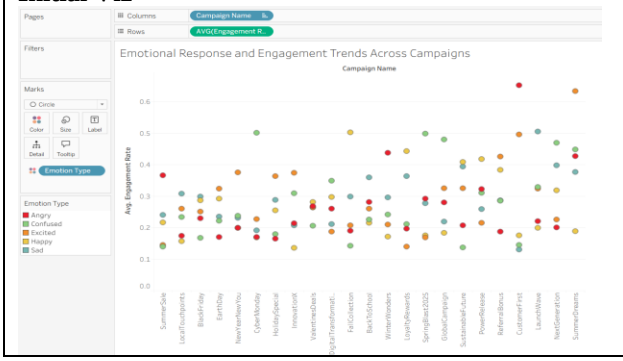
NextGeneration	20.04%	46.89%	22.54%	31.82%	39.77%
SummerDreams	42.69%	44.80%	63.37%	18.79%	37.64%

Chart Type:	Dot plot
Research Question:	Do emotions impact engagement rates across different campaigns?
Fields Used:	• **Calculated Field:** Engagement Rate = (Likes Count + Comments Count + Shares Count) / Impressions • **Rows and Columns:** ○ Columns: Campaign Name ○ Rows: AVG(Engagement Rate) • **Marks / Encodings:** ○ Mark Type: Circle ○ Color: Emotion Type ○ Tooltip: Likes count, Comments count, Impressions count
Enhancements Made:	• Added Viz-in-Tooltip for highlighted points • Label callouts added for outliers • Custom sorting of emotions • Axis adjusted and gridlines formatted for clarity
Business Insight:	Campaigns with emotions like "Happy" or "Excited" generally show higher engagement. For example, NextGenReveal and SummerDreams benefit from positive emotional tones. Outliers like "Angry" reduce engagement. This confirms that emotional design directly influences campaign success.

Initial Viz



Final Viz

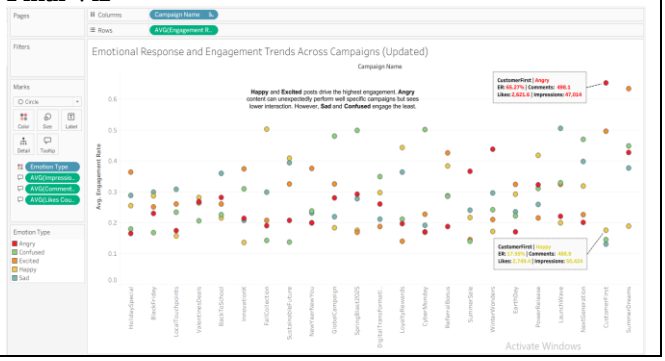
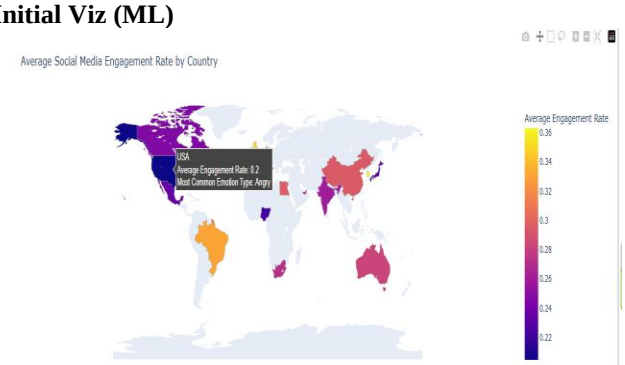
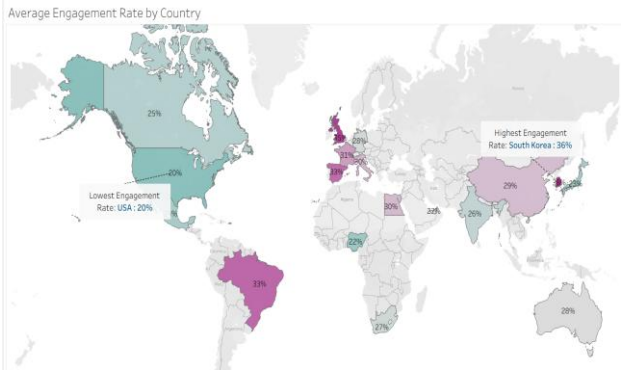


Chart Type:	Choropleth Map
Research Question:	Do engagement rates vary significantly across countries?
Fields Used:	• **Calculated Fields:** Engagement Rate = (Likes Count + Comments Count + Shares Count) / Impressions Country = SPLIT(Location, ", ", 2) • **Rows and Columns:** ○ Columns: Longitude ○ Rows: Latitude • **Marks / Encodings:** ○ Map Style: Filled Map ○ Color: AVG(Engagement Rate) using gradient (light pink to teal) ○ Label: Percentage Engagement Rate per country ○ Detail: Country, ○ Tooltip: Dominant Emotion
Enhancements Made:	• Parsed and cleaned country field from location • Switched to filled map style from point map • Added tooltip with engagement and country summary • Highlighted countries with min/max engagement
Business Insight:	The choropleth map shows higher engagement in South Korea, Canada, and the UK, while countries like Brazil and the USA trail lower. Country-level strategies are essential to improve localization and timing. Focus should be placed on high-performing regions for major launches.

Initial Viz (ML)



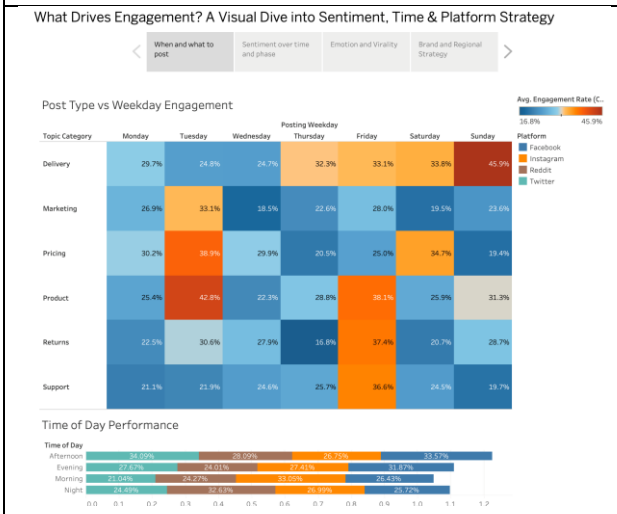
Final Viz (Tableau)



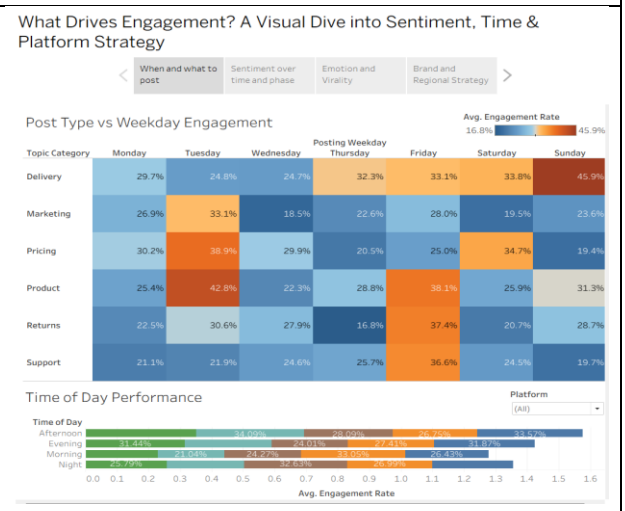
Dashboard Storytelling

When and what to post

Before:



After:



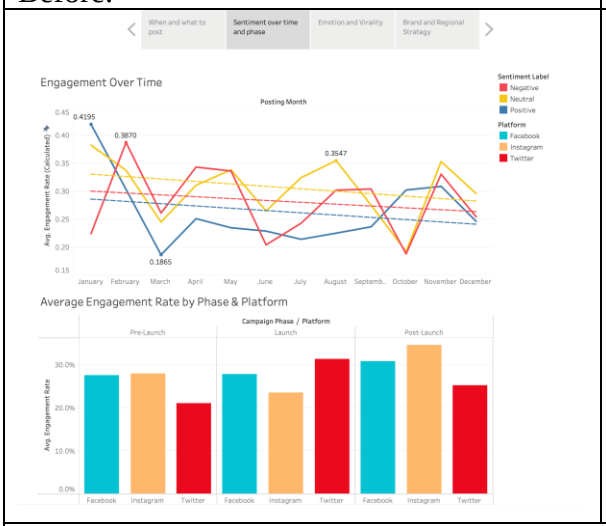
We enhanced clarity in the dashboard storytelling by using floating and filter options to highlight the average engagement rate and platform.

Takeaways:

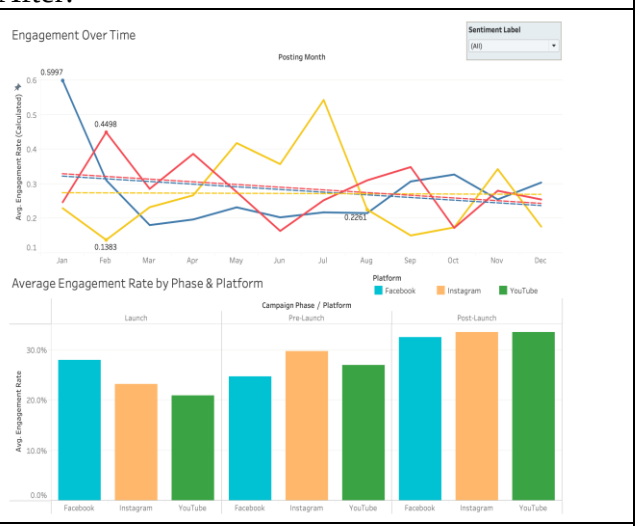
- Shows optimal posting days and content types.
- Helps in scheduling content calendar.
- Reveals peak user engagement hours and platform behavior.

Sentiment Over Time & Phase

Before:



After:



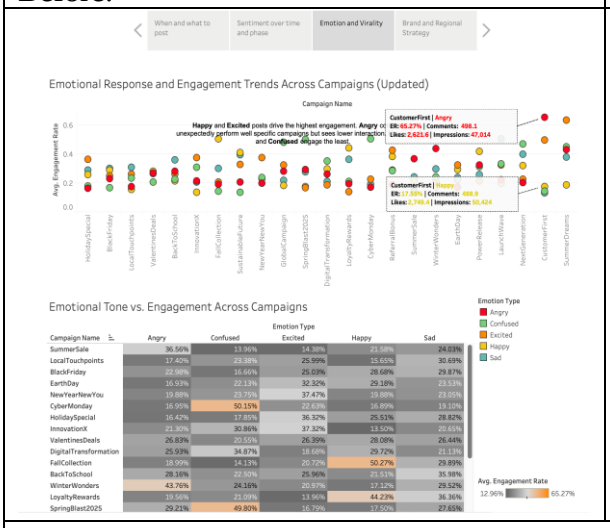
We filtered the platforms to have more dedicated analysis towards more common platforms and their sentiments. This was after our presentation.

Takeaways:

- Demonstrates trendlines of positive, negative, and neutral sentiment impact on engagement over months.
- Reveals for which platform works best at each stage: Pre-launch, Launch, Post-launch.

Emotion and Virality

Before:



After:

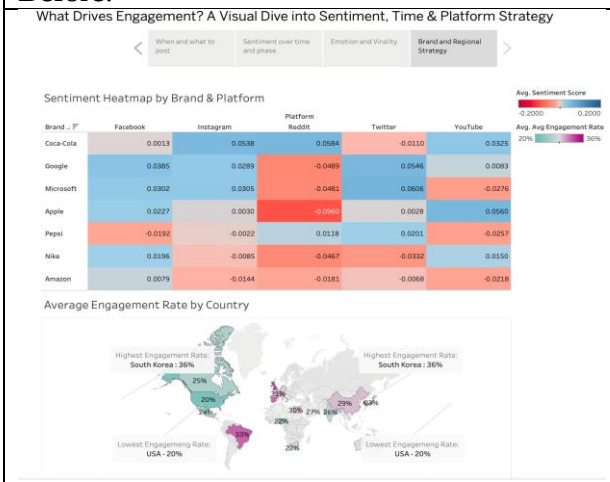
After our presentation, we filtered brand names to focus on the most famous brands and also the three most used platforms: Facebook, Instagram, and YouTube.

Takeaways:

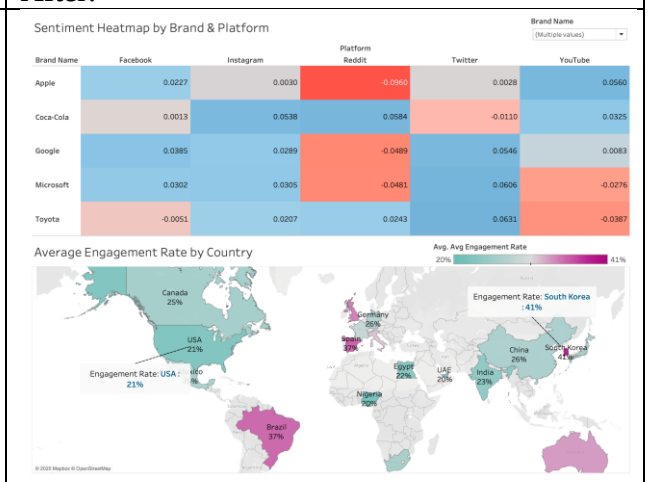
- Highlights which emotions (happy, sad, angry, etc.) influence engagement in which campaigns.
- Visualizes which emotions consistently yield higher engagement across campaigns.

Brand and Regional Strategy

Before:



After:



We filtered the most common brands used worldwide to focus on Apple. Coca Cola, Google, Microsoft, and Toyota. This showed a more concentrated look at relevant brands. After filtering, we can see that South Korea still has the highest engagement rate. Comparatively, the engagement rate for the US has gone up by 1%.

Takeaways:

- Displays sentiment spread by brand and platform (e.g., Twitter has more negative outliers).
- Shows which countries have the highest/lowest engagement.
- Helps tailor region-specific campaigns.

Key Findings

Our analysis determined the key patterns that drove engagement across the different social media platforms we examined. To begin with, two important factors for developing content that captivates the audience are when and what is published: product posts peak in the middle of the week, while delivery related posts perform best on weekends, mostly during the evening hours (6–9 PM). Moreover, these posts generate the highest engagement on Facebook and Instagram, while engagement on Twitter remains relatively stable throughout the day. One of the strongest variables influencing engagement is emotional tone. Positive emotions, such as happiness and excitement, outperformed other emotions in various campaigns, while emotions like sadness and anger showed lower performance.

Furthermore, engagement also changed during the launch and post-launch phases of the campaigns on Facebook and Instagram, showing greater user interaction during these periods. In addition, regional and platform variations were significant: countries like South Korea, Canada and the UK produced the highest engagement rate, and sentiment variation was higher on Reddit and Twitter, platforms that vary more because the audience is guided more by emotions. Overall, the findings suggest that engagement is driven by temporality, emotional design, the behavior of each platform, and the characteristics of the audience by region (Gen AI, 2025)

Use of AI

Open AI and Gen AI helped us craft the initial visualizations and come up with research questions for this project. It helped us craft this report and make initial enhancements to our visualizations (Open AI, 2025)

Challenges Faced

One of our major challenges was the extraction of geographic data from the dataset that posed a significant challenge. The location field contains combined entries, for instance, “Houston, USA” so it's necessary to separate the city and country into different fields to also calculate what emotion is dominant within each country.

Another challenge faced when visualizing the data was that campaigns had an assortment of emotional tones. Multiple tones within a campaign made the visual distribution noisy. To address this issue, we redesigned our visualizations by sorting the emotional distribution and using more distinct color-coded emotional tones.

Additionally, we had an issue with maintaining a consistent scale and color scheme throughout all of the dashboards. Due to comparing charts that had very different value ranges, we needed to complete multiple iterations to ensure the visualizations were uniform.

The creation of a global map also limited the interpretations of the visual dataset. Because not every country has published posts across all of the platforms or campaigns, and due to many areas of the world such as portions of Africa, the Middle East, and much of South America being underrepresented or missing. Therefore, the overall data presented in the global view has a bias toward countries that have a higher number of complete datasets (Open AI, 2025)

Conclusion

This project found those emotions, in particular, happy and excited, tend to spur stronger interactions, and although their effectiveness can vary, they consistently show to be effective, when investigating the drivers of social media engagement. People in different countries react very differently to emotional stimuli, and this difference is seen in the patterns of engagement. The hour of the day, too, plays a major part in shaping the interaction that people have with the content, and the phase of a campaign has a lot of significance.

The use of visualization techniques, including Viz in Tooltip, diverging color scales and custom tooltips, has been instrumental in laying out the findings that would otherwise be hard to spot in the raw data, giving us a much deeper and more intuitive understanding of how the audience is interacting with the content. It is well known to marketers; positive emotions should be the focal point of campaigns and must be adjusted to the regional differences. Data dashboards can aid in real time monitoring to make sure that the tone of the campaign is aligned with its objectives and the expectations of the audience.

It also taught us the importance of thoroughly cleaning and preparing the data. Splitting locations, creating the main emotion characteristics, and clearing up any inconsistencies are all necessary to produce accurate conclusions. Since the dataset was not well-balanced across countries and differed in the use of platforms, we need to take the results into account. This project showcases the capacity of merging data science and emotional understanding to catapult marketing efforts to a higher level, and enables brands to create more engaging, effective, and consumer-focused communications.

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